MILITARY SPECIFICATION

PROPELLANT M30A1
FOR USE IN CHARGE, PROPELLING XM201E2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M30A1 Propellant for use in the XM201E2 Propelling Charge for the 155MM Howitzers, XM198 and M109A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-652 - Propellants Solid for Cannons Requirements and Packing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1171- Acceptance and Description Sheets for Propellants and Explosives
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

FSC: 1375

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DRAWINGS

ARMY

9285409 - Charge, Prop, XM201E2 (White Bag for 155MM How XM198, M109A1

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington,D.C., 20402. Orders for the above publication should cite: 49 CFR 0-199 (latest revision).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M30A1, Type I conforming to the requirements of MIL-STD-652.

3.2 Proving Ground. The propellant, when fired in the XM199 Cannon (with 95% remaining life) and with the weight of projectile (96.0 pounds (lbs)) for which it was manufactured, (M549) shall comply with the following requirements when tested as specified in 4.3.

3.2.1 Assessment. At 70 degrees Fahrenheit (°F) and standard cannon conditions, the propellant shall be capable of charge assessment to a service velocity of 1980 feet per second (fps) at zone 6 and 2245 (fps) at Zone 7 without an associated expected average pressure greater than 34,000 pounds per square inch (psi), or less than 31,000 psi.

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3.2.2 Uniformity of velocity. The standard deviation in velocity for the lot at Zone 7 at 70°F shall not exceed 6.0 fps.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 58,800 psi any any operating temperature (-50°F to +145°F) when fired in a standard cannon.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inpsection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109, in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of batches of product inspected (shall be made available upon request by the contracting officer).

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c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer)

d. Drawing, specification number, and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contracting documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet SMU Form 1060 containing the applicable data (see 6.8)

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots or one or more batches of M30A1 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M30A1 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

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4.2.2 Testing

4.2.2.1 Materials (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.2.2.2 Proving ground (see 3.2). Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652.

4.2.2.2.1 Assessment (see 3.2.1) Major defect, Code No. 01001 (see 6.4) The lot shall be rejected if the assessed charge weight of propellant has an expected average pressure greater than 34,000 psi or less than 31,000 psi.

4.2.2.2.2 Velocity Uniformity (70°F) (See 3.2.2) Major defect, Code No. 02001. If the product of the factor (see 6.5) times the velocity standard deviation of the rounds tested in accordance with 4.3.3 exceeds the applicable requirement the lot shall be rejected. For initial production lots a ten round sample shall be tested (see 6.6)

4.2.2.2.3 Individual pressure (see 3.2.3). If any individual pressure observed during uniformity firings at any operating temperature after correction for difference between as fired and recommended service charge weight exceeds 58,800 psi, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Proving ground tests. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.3.2 Charge assessment

4.3.2.1 Charge Slope Method. Three or more groups of charges loaded with selected charge weights of propellant (weights seleoted should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which

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will yield the prescribed service velocity (para 3.2.1) This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.3.3). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.3.2.2 Constant Slope Method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.3.2.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.3.3 Uniformity series

4.3.3.1 Production

4.3.3.1.1 145°F. Ten rounds of XM201E2 propelling charges shall be assembled with the assessed charge of propellant, conditioned to 145°F (as indicated by a thermocoupled control round (rnd)) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight).

4.3.3.1.2 70°F. Twenty rounds of XM201E2 charges (10 each at Zone 6 and 7) shall be assembled with the assessed charge of propellant, conditioned to 70°F (as indicated by thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means, as well as the standard deviations associated with these parameters

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shall be calculated. All results shall be corrected to standard shell weight (96.0 lbs) and standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions) and to recommended charge weight conditions.

4.3.3.1.3 -50°F. Ten rounds of XM201E2 propelling charges shall be assembled with the assessed charge of propellant, conditioned to -50°F (as indicated by a thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight).

4.3.3.2 Initial Production Lots (For Informational purposes only).Ten charges each at -50°F, +70°F +145°F (total 30 rds) should be fired for Pressure vs Time in an instrumented XM199 Cannon where Piezo pressure vs time and ignition delay are recorded in the breech spindle and at a point 35.5 inch RFT (tapped in cannon tube). Muzzle velocity and pressures should also be recorded.

4.3.4 Refiring. If for any reason the proving ground considers that test conditions have detrimentally affected the test result, additional charges as required shall be tested.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)

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c. Process. Details of the manufacturing process and the equipment used by the Contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-ADEP-2, Dover, New Jersey 07801.

6.4 Inspection code number. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.6 Initial production lots. The initial production lots represent the first three consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to initial production plan.

6.7 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.8 Submission of SMU Form 1060. Information contained in propellant description sheets, SMU Form 1060, need not be resubmitted in the form of certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.2).

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6.9 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, N.J. ATTN: SARPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.10 Test charges shall be assembled in accordance with drawing 9285409.

6.11 Proving ground test summary

PRODUCTION

Temp (°F)	Sample Size	Vel (fps) (assessed)	Lot Std Dev of Vel (fps)	Perm Ind Max Pres (psi)	Avg. Pres. Max (psi) min	max
-50 (Zone 7)	10	n/a	n/a	58,800	n/a	n/a
70 (Zone 6)	10	1980	n/a	58,800	n/a	n/a
70 (Zone 7)	10	2245	6.0	58,800	31,000	34,000
145 (Zone 7)	10	n/a	n/a	58,800	n/a	n/a

PRESSURE VS TIME EVALUATION

-50 (Zone 7)	10
70 (Zone 7)	10
145 (Zone 7)	10

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A-94

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